October 8, 2025

1 Lab3 Report

1.1 Overview

The machine follows a defined interaction sequence: **attendance → greeting → two subtle custom gestures**
→ one not subtle gesture → farewell.

Pipeline:

```
AttendUser → Greet → ThinkingGesture → EyeRollGesture →  SurpriseGesture → Farewell →
```

Each state invokes an asynchronous actor and transitions upon completion, logging progress and failing safely.

1.2 Design for subtlety

The initial gestures were intentionally designed to appear micro-expressive, focusing on small movements.

ThinkingGesture. Executed with `blocking = false` and minimal amplitude parameters to evoke a subtle "thinking" impression:

- **Head posture:** slight neck tilt ($-15°$) and gentle pan ($10 \to 12°$)
- **Facial adjustment:** right brow raise ($3.0 \to 0.7$) and light eye squint ($\sim 1.0 \to 0.5$)
- **Reset:** quick neutral frame at $t = 2.0\,\text{s}$

EyeRollGesture. This gesture preserves subtlety but adds mild expressiveness:

- gradual upward eye movement (`LOOK UP 6→17`)
- marked holding frame at $t = 1.1\,\text{s}$ for
- smooth reset at $t = 2.5\,\text{s}$
- light blinks (intensity ≈ 0.6) to avoid a slapstick effect

1.3 Breaking the Subtlety

The final gesture, `SurprisedWithAudioGesture`, intentionally disrupts the micro-gesture pattern to achieve comic through exaggerated movements.

- **Facial expression:** brows up, neck-roll oscillation, and open smile, timed at 0.0, 0.15, and 0.4 s before reset.
- **Speech:** paired with an exaggerated vocal outburst:

```
fhSay("WHAT IS GOING ON? ARE YOU OKAY?")
```

The exaggerated motion and timing provide humorous payoff after the preceding subtle buildup.

1.4 Speech–gesture combination

Subtle gestures are paired with mild speech lines (e.g., "Let me think..."), while the exaggerated gesture is coupled with a strong verbal reaction. An attempt was made to include a goodbye kiss gesture, but I could not find it in the documentation.